1. Core Correlation: Point 7

> **Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis**

- **Analysis:** This is the primary requirement of the task. To solve the problem, a student must **decompose** the 18 km round trip into two distinct parts: the 9 km ascent and the 9 km descent. They must further simplify "twice that speed" into a numerical value (3.0 km/h) before any final calculation can occur.

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2. High Correlation: Point 4

> **Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation**

- **Analysis:** The student must **decode** the narrative text to extract a mathematical timeline. This involves making sense of the constraint "return by 8 pm" and the phrase "taking into account meal breaks" to understand that the given speeds are already net averages, eliminating the need to add extra time for rests.

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3. High Correlation: Point 2 & Point 6

> **Identify the key variables in a model / Identify and describe the mathematical aspects... including identifying the significant variables**

- **Analysis:** Success depends on **identifying significant variables**:
  - **Distance:** 9 km (up) and 9 km (down).
  - **Speed:** 1.5 km/h (up) and 3.0 km/h (down).
  - **Constraint:** Latest return time of 8:00 pm.
  - **Goal Variable:** Start time T.

4. Moderate Correlation: Point 9

> **Translate a problem into a standard mathematical representation or algorithm**

- **Analysis:** The student must **translate** the physical situation into a standard algorithm: $Time = \frac{Distance}{Speed}$. The final representation involves a subtraction algorithm (8:00 pm minus total travel time) to find the starting point.

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5. Moderate Correlation: Point 12

> **Identify the constraints, assumptions simplifications in a mathematical model**

- **Analysis:** The problem requires identifying a hard **constraint** (8:00 pm) and an **assumption** provided in the text (that the estimated speeds are constant averages that already include breaks). Recognizing these allows the student to build a linear model of time versus distance.

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6. Lower Correlation: Point 5

> **Recognise mathematical structure (including regularities, relationships, and patterns)**

- **Analysis:** The student recognizes the **inverse relationship** between speed and time (higher speed down means less time spent) and the additive structure of the total trip time.

1. 核心关联项：第 7 条

> **Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis** (简化或分解情境或问题，使其适合进行数学分析)

- **分析：** 这是解决此任务的主要要求。为了解决问题，学生必须将 18 公里的往返行程**分解**为两个不同的部分：9 公里上坡和 9 公里下坡。在进行任何最终计算之前，他们必须进一步将“两倍的速度”简化为一个数值（3.0 公里/小时）。

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2. 高度关联项：第 4 条

> **Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation** (阅读、解码陈述、问题、任务、物体或图像，并理解其含义，以创建情境模型)

- **分析：** 学生必须**解码**叙述性文本，以提取数学时间线。这包括理解“晚上 8 点前返回”的限制条件，以及“考虑用餐休息时间”这句话，从而理解给定的速度已经是平均净速度，无需额外添加休息时间。

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3. 高度关联项：第 2 条 和 第 6 条

> **Identify the key variables in a model / Identify and describe the mathematical aspects... including identifying the significant variables** (识别模型中的关键变量 / 识别和描述现实世界问题情境的数学方面.....包括识别重要变量)

- **分析：** 成功取决于**识别重要变量**：
  - **距离：** 9 公里（上坡）和 9 公里（下坡）。
  - **速度：** 1.5 公里/小时（上坡）和 3.0 公里/小时（下坡）。
  - **限制：** 最晚返回时间为晚上 8:00。
  - **目标变量：** 开始时间 T。

4. 中度关联项：第 9 条

> **Translate a problem into a standard mathematical representation or algorithm** (将问题转化为标准的数学表示或算法)

- **分析：** 学生必须将物理情境**转化**为标准算法：$\text{时间} = \frac{\text{距离}}{\text{速度}}$。最终的表示涉及减法算法（晚上 8:00 减去总旅行时间）以找到起点时间。

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5. 中度关联项：第 12 条

> **Identify the constraints, assumptions simplifications in a mathematical model** (识别数学模型中的限制、假设和简化)

- **分析：** 问题要求识别一个硬性**限制**（晚上 8:00）和文本中提供的一个**假设**（估计速度是已经包含休息时间的恒定平均值）。识别这些使学生能够建立时间和距离的线性模型。

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6. 较低关联项：第 5 条

> **Recognise mathematical structure (including regularities, relationships, and patterns)** (识别问题或情境中的数学结构（包括规律性、关系和模式）)

- **分析：** 学生**识别**速度与时间之间的**反比关系**（下坡速度更快意味着花费的时间更少）以及总行程时间的加法结构。